

RESEARCH



LesionScanNet: dual-path convolutional neural network for acute appendicitis diagnosis

Muhab Hariri¹, Ahmet Aydın², Osman Sibiç³, Erkan Somuncu⁴, Serhan Yılmaz⁵, Süleyman Sönmez⁶ and Ercan Avşar^{7*} 

Abstract

Acute appendicitis is an abrupt inflammation of the appendix, which causes symptoms such as abdominal pain, vomiting, and fever. Computed tomography (CT) is a useful tool in accurate diagnosis of acute appendicitis; however, it causes challenges due to factors such as the anatomical structure of the colon and localization of the appendix in CT images. In this paper, a novel Convolutional Neural Network model, namely, LesionScanNet for the computer-aided detection of acute appendicitis has been proposed. For this purpose, a dataset of 2400 CT scan images was collected by the Department of General Surgery at Kanuni Sultan Süleyman Research and Training Hospital, Istanbul, Turkey. LesionScanNet is a lightweight model with 765 K parameters and includes multiple DualKernel blocks, where each block contains a convolution, expansion, separable convolution layers, and skip connections. The DualKernel blocks work with two paths of input image processing, one of which uses 3×3 filters, and the other path encompasses 1×1 filters. It has been demonstrated that the LesionScanNet model has an accuracy score of 99% on the test set, a value that is greater than the performance of the benchmark deep learning models. In addition, the generalization ability of the LesionScanNet model has been demonstrated on a chest X-ray image dataset for pneumonia and COVID-19 detection. In conclusion, LesionScanNet is a lightweight and robust network achieving superior performance with smaller number of parameters and its usage can be extended to other medical application domains.

Keywords: Acute appendicitis, CNN, Deep learning, Medical image classification, Disease detection

Introduction

Appendicitis is defined as the inflammation of the appendix, and it is an urgent medical emergency situation. This condition often manifests as acute abdominal pain, which typically requires immediate appendectomy. The diagnosis of acute appendicitis without any medical imaging modalities is difficult because the symptoms of acute appendicitis are similar to those observed in other pathological conditions. Computed Tomography (CT) scanning of the abdomen and pelvis enables visualization of appendix and hence provides a useful tool for its state evaluation. However, there are still notable challenges associated with the diagnostic process when CT imaging is utilized. The complexities in the anatomical structure

of the colon together with the anatomic location of the appendix are the two major factors introducing challenges on the identification of the inflamed appendix and hence decreasing the likelihood of timely diagnosis. Therefore, improving clinical decision-making is critical in enhancing the diagnosis and treatment of acute appendicitis. Considering these limitations, utilization of computational algorithms, such as Convolutional Neural Networks (CNNs), can be useful to automatize diagnostic accuracy and expedite the identification of acute appendicitis.

The incorporation of deep learning methodologies for making diagnosis from medical images is a research area with numerous applications. For example, CNNs have been applied in the detection of various pathologies such as cancer [1], respiratory complications [2, 3], and gastrointestinal problems [4]. However, identification of acute appendicitis using CNN models comes with some unique challenges. In the case of CT scans, even if lesions are

*Correspondence: erca@aqu.dtu.dk

⁷ Section for Fisheries Technology, Institute of Aquatic Resources, DTU Aqua, Technical University of Denmark, 9850 Hirtshals, Denmark
Full list of author information is available at the end of the article

present and indicate acute appendicitis, they tend to be very small and barely visible, making it difficult to detect them using both manual and automated approaches including CNN-based models [5]. In addition, complicating factors such as different qualities of CT scans, anatomic characteristics between patients, and stages of pain severity are all able to alter the appearance of the lesion, thereby posing issues for consistent performance measures by CNN models.

In light of the aforementioned difficulties, this study proposes a novel CNN architecture, namely LesionScanNet, to address the complexities associated with extracting the features of the lesions within CT images particularly of acute appendicitis. In particular, LesionScanNet comprises five DualKernel Blocks (DKBs) that consist of two separate processing paths with different kernel sizes. Each path in a DKB structure comes with a sequence of a convolution layer, expansions, separable convolution layer [6], and skip connections. The outputs of each path are merged through concatenation layer that is followed by a max-pooling operation. Each DKB reduces the width and height of the feature maps while increasing the number of channels. LesionScanNet has been tested on a dataset of CT images belonging to the abdominal region of 800 individuals. The performance of the LesionScanNet has been compared with the nine generic CNN models available in the literature, which are EfficientNetB0, EfficientNetB2, InceptionV3, InceptionResNetV2, MobileNetV2, NASNetMobile, ResNetV2_101, VGG16, VGG19. It has been shown that LesionScanNet has relatively lower number of parameters than the generic models and higher classification performance on the dataset.

Considering the advantages of the LesionScanNet, the contributions of this study can be listed as follows:

- A novel CNN architecture, LesionScanNet, has been proposed to detect lesions in medical images, specifically those presenting acute appendicitis. In particular, DKB structure has been developed for an efficient feature extraction while keeping the amount of model parameters low.
- LesionScanNet provides remarkably high classification performance, surpassing other generic CNN models in detection of diseases using different types of medical images (i.e. acute appendicitis from CT and pneumonia from radiography images), indicating its wide applicability in different clinical scenarios.
- The proposed model has only 765 K parameters, a value much lower than generic CNN models. This feature makes it suitable to be used in hardware with limited computational capability.

The structure of the paper is as follows: Sect. “[Related work](#)” presents the relevant literature, Sect. 3 introduces the dataset, data processing, models involved in this study, and performance metrics, Sect. “[Results](#)” presents the results of the models, ablation study, and generalization study, Sect. “[Discussion](#)” provides discussions on the results and the paper is concluded in Sect. “[Conclusion](#)”.

Related work

In the field of appendicitis detection, there are many studies that have been conducted highlighting the effectiveness of machine learning algorithms either through deep learning or traditional methodologies. Deep learning-based approaches typically use unstructured data such as CT images to develop CNN models. An example is AppendiXNet, a three-dimensional (3D) CNN model for computer-assisted detection of acute appendicitis cases [7]. A small dataset of 438 contrast-enhanced CT images was used in the experiments and video-pretraining was shown to increase the area-under-curve (AUC) value from 0.724 to 0.810. Another 3D CNN model was proposed by Park et al. for binary classification of CT images as appendicitis or normal [8]. The 3D CNN model consisted of several convolutional, max-pooling, and fully connected layers and was validated using 667 CT images acquired from 215 patients. The proposed model obtained a maximum accuracy of 97.5% for diagnosing acute appendicitis among patients in an emergency condition, which reflects the efficiency of the CNN. To address the limitations related with 3D CNN approach, superposition of the red, green, and blue channels in the CT images was proposed to train CNN models by Lee et al. [9]. The study used 2782 axial-slice CT images from 500 people to train four variants of EfficientNet model and it was shown that the EfficientNetB4 achieved the highest performance scores of 91.98% and 95.35% for accuracy and recall, respectively. In addition to supervised learning methods mentioned above, a semi-supervised representation learning approach was proposed to address the problems associated with incomplete data [10]. The study used ultrasound images to detect pediatric appendicitis and developed a multiview concept bottleneck model with a semi-supervised extension. The dataset contained ultrasound images from 579 patients and achieved an AUC score of 0.80. Reinforcement Learning (RL) was used as an option to reduce the search space for diagnosis of acute appendicitis by Al et al. [5]. Their model involved combination of Fully Convolutional Network (FCN) and Region of Low-Entropy (RLE) to determine the ultimate probability of acute appendicitis in CT images. The evaluation was performed on a dataset of 319 abdominal CT volumes, 116 of which were

diagnosed to have acute appendicitis and an AUC score of 0.96 was achieved.

On the other hand, traditional machine learning methodologies are still being used in recent years for diagnosis of acute appendicitis. In general, these methods receive structured data such as clinical examinations, laboratory findings, and medical history of the patients for model training. Some examples include utilization of Random Forest method for appendicitis diagnosis [11, 12] and severity of pediatric appendicitis [13]. Decision trees and gradient boosted decision trees are two alternatives found to be useful for similar applications [14, 15]. In addition, there are studies involving training of artificial neural networks (ANN) [16, 17], and support vector machines (SVM) together with k-nearest neighbors (KNN) [18] models using clinical information of the patients. Despite the presence of traditional approaches in the literature, they have not been discussed in detail here since they typically use a different approach with other types of data.

Based on the observations on the available studies about detection of acute appendicitis, it is possible to claim that there are still possibilities for improvement in this application domain. Furthermore, none of these studies focuses on simultaneously improving the model accuracy and reducing its computational cost. To address this gap, LesionScanNet has been proposed in this study, which is relatively a lightweight model that can extract representative visual features with less amount of model parameters.

Materials and methods

Dataset

The dataset includes 2400 CT slice images collected from 800 individuals between the dates January 1st, 2019 and June 30th, 2023 at the Department of General Surgery at İstanbul Kanuni Sultan Süleyman Training and Research Hospital, Turkey. The dataset has undergone a meticulous processing for compliance with regulations, with great emphasis on protection of patient confidentiality. All the images in the dataset have been anonymized and it has been made sure that none of the individuals are under 18 years of age and have an urgent pathology. The demographic distribution includes both genders, with males comprising approximately 67% and females 33% of the dataset. Specifically, the acute appendicitis group shows a slightly higher male prevalence (66.5%) compared to the healthy group (67.25% male), with similar gender ratios across groups. Age distribution is varied, spanning from 18 to over 86 years, with a concentration in the 18–25 and 26–35 age groups. Ethical committee approval has been granted for the utilization of the dataset in research endeavors. All the CT images were

manually inspected by general surgery specialists to label them as either “healthy” or “acute appendicitis”. The dataset has been formed by including three images for each of the 800 individuals, with half being healthy and the other half diagnosed with acute appendicitis. In other words, 1200 images in the dataset belong to 400 healthy individuals and the remaining images belong to 400 individuals with acute appendicitis. The CT images were collected during the diagnostic process, with three different slices for each acute appendicitis case showing the condition and three slices without disease findings for the healthy group. All images were saved in JPEG format with a resolution of 1164×734 pixels. Figure 1 illustrates three sample CT scans of a patient with acute appendicitis (left column) and a healthy case (right column).

The dataset was randomly split into three subsets training, validation, and test sets with a distribution proportion of 75%, 12.5% and 12.5%, respectively and it has been made sure that images from the same individual are in the same subset. Some image augmentation techniques have been applied to the training set to capture the potential deviations in the CT images due to the posture of the individual and eventually reduce the risk of overfitting. The augmentation techniques along with their related parameters are presented in Table 1.

Data preprocessing

Enhancement of the medical images as a preprocessing step is known to have positive impact on the performance of deep learning models [19, 20]. Therefore, two image enhancement methods have been applied to the entire dataset before proceeding with the model training and evaluation. Image sharpening is followed by Contrast Limited Adaptive Histogram Equalization (CLAHE) as the preprocessing step to improve the CT image quality and eventually obtain better feature representations.

Image sharpening highlights the edges in the image by convolving it with a specialized kernel shown in Eq. 1. The aim of CLAHE is to improve the image contrast at regional levels. This process is done by dividing the image into smaller tiles and then applying histogram equalization to each of them, thereby eliminating over-amplification of noise. Sharpening and CLAHE can be used collaboratively in medical image processing because sharpening defines fine structures such as lesions, while CLAHE increases the contrast between different tissue densities. This two-step methodology not only helps radiologists in visual diagnostics but also increases the effectiveness of automated image analysis algorithms, which allows for making more precise and reliable medical assessments.

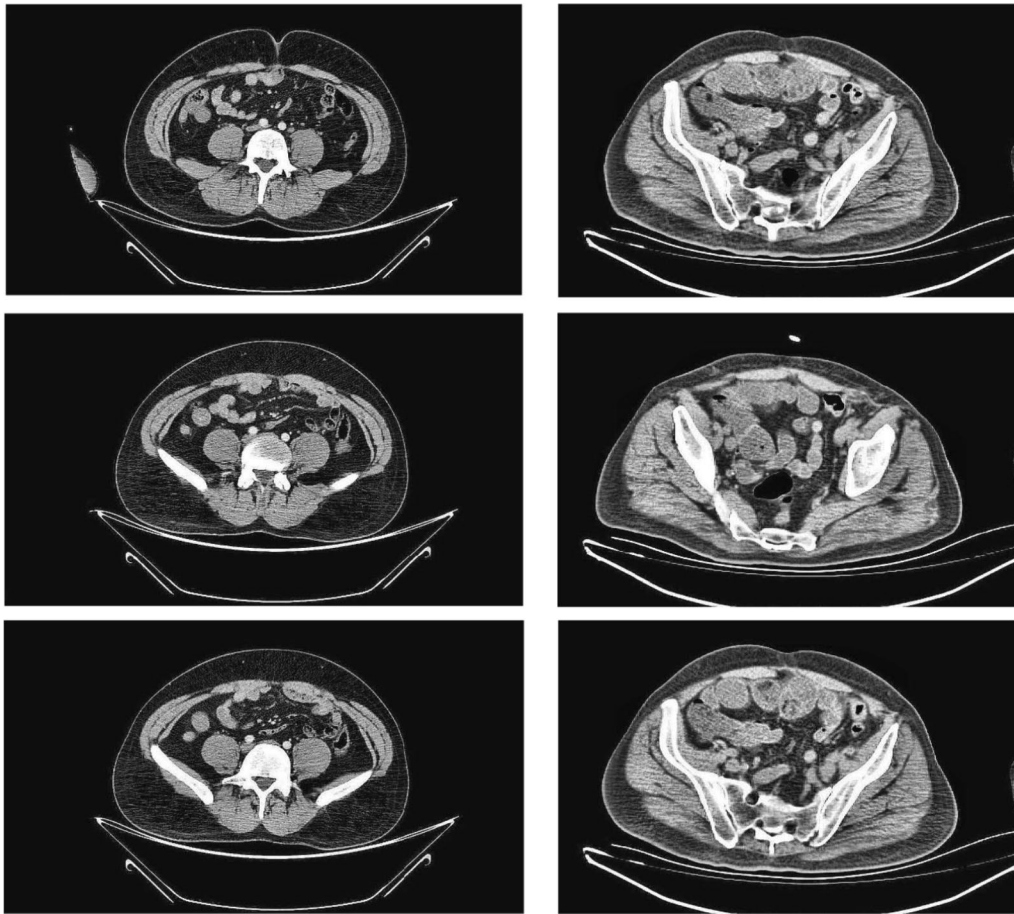


Fig. 1 Triple samples of acute appendicitis (left column) and healthy (right column) CT images representing two cases

Table 1 The data augmentation techniques applied to the data, along with their parameters

Technique	Intensity range
Rotation Range	$[-10, 10]$ degrees
Width Shift Range	$[-0.1, 0.1]$ (10% of total width)
Height Shift Range	$[-0.1, 0.1]$ (10% of total height)
Shear Range	$[-0.1, 0.1]$ (10% shear)
Zoom Range	$[-0.1, 0.1]$ (10% zoom in or out)
Horizontal Flip	[True, False]

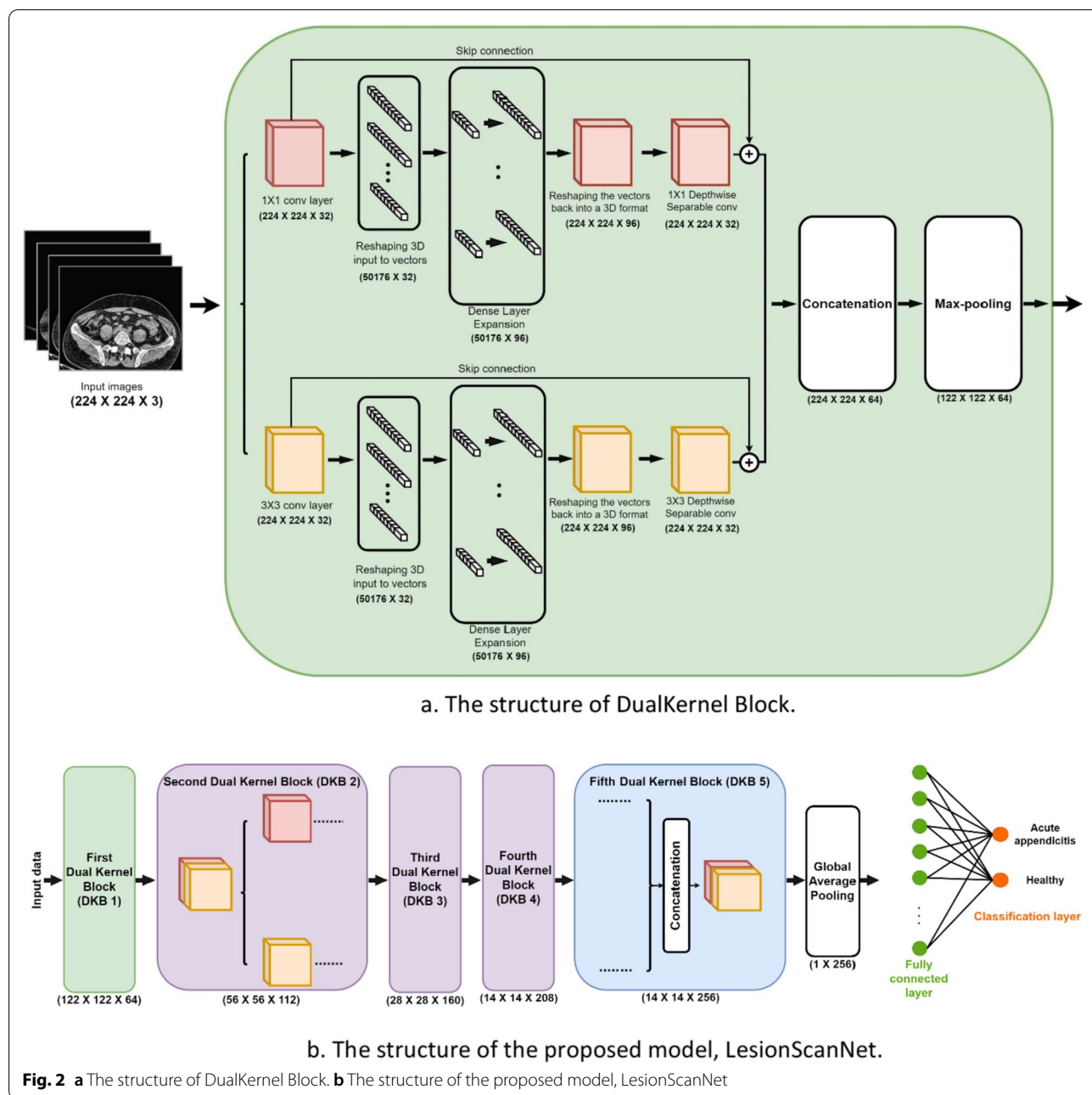
$$\text{Kernel} = \begin{pmatrix} -1 & -1 & -1 \\ -1 & 9 & -1 \\ -1 & -1 & -1 \end{pmatrix} \quad (1)$$

The proposed LesionScanNet model

The proposed CNN architecture, namely LesionScanNet, has been designed for the detection of acute appendicitis

cases in CT images. The structure of LesionScanNet incorporates five cascaded DKBs that contain two parallel pathways for processing the input tensors. Five DKBs are used to balance spatial information retention and dimensionality reduction, as fewer blocks increase parameter count, while more blocks reduce prediction accuracy. Every path consists of a convolutional layer, an expansion layer, depthwise separable convolutions, and an additive skip connection. The outputs of each processing pathway are combined via concatenation operation. All the DKBs until the last block have a 2×2 max pooling operation after concatenation while the final DKB has a global average pooling as the final step. Figure 2a and b show the structures of a generic DKB and the proposed LesionScanNet model, respectively. In Fig. 2, the expressions in parentheses denote the output dimensions of the blocks.

Initially, LesionScanNet takes the input images with a resolution of $224 \times 224 \times 3$. The input is passed through two processing branches inside the first DualKernel Block (DKB 1), which are the core elements of feature



extraction. One branch uses convolution layers with 1×1 kernels mainly for manipulating features at the channel level and enriching the feature representation power of the network. This is particularly important in medical imaging where specific channels may contain the hidden diagnostic information. Through channel-wise processing, this branch also preferentially amplifies or diminishes certain aspects of the input to enhance the appearance of subtle diagnostic markers such as lesions. On the other hand, the branch with 3×3 kernels focuses

mainly on spatial feature extraction which is essential for identifying tissue abnormalities. The 3×3 kernel size is a compromise between the capturing of important spatial patterns and preserving fine detail which is critical for accurate lesion detection in medical images. In particular, it balances the need for capturing fine spatial details and maintaining computational efficiency. Furthermore, deep hierarchical learning helps the model learn both low and high-level spatial patterns by stacking multiple 3×3 kernels, improving its ability to identify various lesion sizes

and shapes, which is crucial for accurate diagnosis. Accurate diagnosis of different anatomical structures requires this layered feature extraction to identify objects of varying sizes.

The input in both branches starts with a convolutional layer. Then, the output of this convolutional layer is reshaped into a sequence of vectors where the matrix in each channel is reshaped into a vector. This operation can be considered as a preparation for the next step, i.e., the expansion layer. An expansion layer with factor of 3 was used to enable the network to represent a broader set of channel features and capture more complex patterns and subtleties. This enlargement of the feature space, through the ReLU activation function, enhances the ability of the model to identify nonlinear relationships within the channel features which is critical for deep learning models for complex decision-making tasks. Following the expansion, the output is reshaped again to prepare it for the depthwise separable convolution process, which efficiently executes filtering and channel combination. The depthwise separable convolution layer works on returning the number of filters to their original count after expanding them by the expansion step with a factor of 3. This two-step process (depthwise followed by pointwise convolution) is much more parameter-efficient than standard convolutions. This advantage is substantial for medical imaging where high-resolution data requires significant computational resources. This efficiency helps in building deeper networks that require less computational load and lower memory usage. This reduces the risk of overfitting, which is particularly important given the complex nature of medical imaging. The network includes additive skip connections between the first convolutions and the output of depthwise separable convolutions. This approach helps in retaining the feature information across the network in addition to handling the problem of vanishing gradient. It ensures a straight flow of gradient during the backpropagation, making it easier to train deeper networks while preserving some spatial information that might be lost during the expansion. The outputs of these dual pathways are then concatenated and passed through a max pooling layer. Max pooling reduces the spatial dimensions of input feature maps, thus reducing both parameters and computations in the network. Max Pooling uses a 2×2 window to select dominant features and eliminate the less significant details.

The following DualKernel blocks (DKB 2, DKB 3, DKB 4, and DKB 5) initially separate the feature maps output by the previous DKB such that each set of feature maps follows its respective branch. In other words, the output from the 1×1 kernel size branch in the previous block passes through the same 1×1 kernel size branch in the next block.

Similarly, the output from the 3×3 kernel size branch in the previous block goes through the same 3×3 kernel size branch in the next block.

The max pooling layer is replaced by global average pooling in the final DKB to aggregate the spatial features into a channel-wise vector. This vector is then passed through one fully connected layer with 64 hidden units followed by ReLU activation, allowing the network to capture more complex non-linear feature relationships. The presence of the dense layer provides architectural flexibility, which is essential to the learning of complex functions. Table 2 provides details about the number of filters used in each block.

Transfer learning

In order to benchmark the performance of the LesionScanNet model, a variety of pre-trained models utilizing transfer learning methods were used. Transfer learning, which is an essential technique in machine learning, relies on the usage of the weights learned from pre-trained models on massive datasets like ImageNet and fine-tuning them for new, sometimes narrower purposes. This makes this method especially beneficial in cases of limited data or when training a model from scratch is too computationally expensive.

For this purpose, the images in the dataset (Sect. 3.1) were pre-processed as explained in Sect. 3.2 and nine benchmark CNN models (EfficientNet B0 and B2 [21], InceptionV3 [22], InceptionResNetV2 [23], MobileNetV2 [24], NASNetMobile [25], ResNetV2_101 [26], VGG16 and VGG19 [27]) pretrained on ImageNet dataset have been fine-tuned using them [28]. When applying transfer learning, the classification layers of the benchmark models were adjusted to have two units (i.e., the number of classes in the dataset) together with a Softmax activation function. The benchmark models differ in input image dimensions, layer complexity, and total number of parameters. These architectures are listed and compared in Table 3 together with LesionScanNet model.

Performance evaluation metrics

To evaluate the performance of the proposed and benchmark models, four evaluation metrics have been used. First, the confusion matrix is generated using the predictions on the test (Table 4). Since the aim is to detect acute appendicitis cases, the samples with the disease have been considered as the positive class in the confusion matrix.

Table 2 Numbers of filters used in the proposed LesionScanNet model

	DKB 1	DKB 2	DKB 3	DKB 4	DKB 5
Number of filters	32	56	80	104	128

Table 3 Comparison between the proposed model and transfer learning models regarding input image dimensions and number of parameters

	Input size	Number of parameters
EfficientNet B0	224 × 224 × 3	4 M
EfficientNet B2	260 × 260 × 3	7.7 M
InceptionV3	299 × 299 × 3	21.8 M
InceptionResNetV2	299 × 299 × 3	54.3 M
MobileNetV2	224 × 224 × 3	2.2 M
NASNetMobile	224 × 224 × 3	4.2 M
ResNetV2_101	224 × 224 × 3	44.5 M
VGG16	224 × 224 × 3	134.2 M
VGG19	224 × 224 × 3	139.5 M
LesionScanNet	224 × 224 × 3	0.765 M

Best values are given in bold

Table 4 The confusion matrix

	Predicted	
	Acute appendicitis	Healthy
Actual		
Acute appendicitis	True Positive (TP)	False Negative (FN)
Healthy	False Positive (FP)	True Negative (TN)

Using the entries of the confusion matrix, accuracy, precision, recall, and F1-score values are calculated (Eqs. 2–5). Accuracy is a measure of correct prediction rate and provides an overall performance evaluation. More detailed performance assessment is enabled by calculating the other metrics where precision and recall considers false positive and false negative predictions, respectively. The F1-score provides a balanced measure between precision and recall values through their harmonic mean. These metrics provide a detailed assessment of the models involved in the experiments for medical image classification.

$$Accuracy = \frac{TP + TN}{Totalnumberofcases} \times 100\% \quad (2)$$

$$Precision = \frac{TP}{TP + FP} \quad (3)$$

$$Recall = \frac{TP}{TP + FN} \quad (4)$$

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (5)$$

Training settings and hardware

Adam optimizer [29] with a learning rate of 0.0005 was used for training the proposed model. This value was chosen empirically, meaning that the experiments were carried out using three different learning rates: 0.001, 0.0005, and 0.00025. The corresponding results indicated that reducing Adam optimizer's default learning rate from 0.001 to 0.0005 improved prediction accuracy. However, further reducing the learning rate to 0.00025 did not result in additional accuracy gains. Therefore, a learning rate of 0.0005 was selected as the optimal value. The exponential decay rate (β_1 and β_2) for the first and second moment estimates were set to default values of 0.9 and 0.999, respectively. In addition, default epsilon value of 10^{-7} was included to improve the numerical stability. The model was trained for 200 epochs and the weights corresponding to the best validation results were used to evaluate the performance on the test set. Python (Version 3.10.9) was used for implementation with the aid of Tensorflow (Version 2.10.0) framework. The training process was performed on a pair of Nvidia GeForce RTX 3060 GPUs. However, the inference time was measured using only one Nvidia GeForce RTX 3060 GPU, with a batch size of 1.

Results

LesionScanNet and transfer learning results

The results of the study enable performance comparison of different deep learning models in discrimination between 'Acute appendicitis' and 'Healthy' samples. The model training was initialized after applying the preprocessing (Sect. 3.2) and augmentation (Sect. 3.1) steps. The classification performances of the trained models were assessed using the evaluation metrics (Sect. "Performance evaluation metrics") on the test set. Table 5 shows the results of the benchmark models and the proposed model LesionScanNet.

Among the benchmark models, the highest classification accuracy has been achieved by NASNetMobile with the lowest amount of false negative predictions. As a result, the best values for recall and F1-scores are also associated with the NASNetMobile model. On the other hand, the highest precision value has been achieved by the MobileNetV2 model as a result of the lowest amount of false positive predictions. NASNetMobile is followed by EfficientNet B0, Inception-ResNetV2, MobileNetV2, and EfficientNet B2 all of which have accuracy values above 88%. The accuracy scores of the remaining four models, namely, InceptionV3, ResNetV2_101, VGG16, and VGG19 are below 80% indicating a poor discrimination performance between the classes.

Table 5 Prediction results of the LesionScanNet and benchmark models, along with computational cost metrics. Best values are given in bold

Model	Precision	Recall	F1-Score	Accuracy (%)	Confusion matrix		Number of parameters (millions)	Inference time per image (ms)
EfficientNet B0	0.882	0.900	0.891	89.00	135 18	15 132	4	15.02
EfficientNet B2	0.914	0.847	0.879	88.33	127 12	23 138	7.7	18.24
InceptionV3	0.880	0.633	0.736	77.33	95 13	55 137	21.8	18.19
InceptionResNetV2	0.842	0.960	0.897	89.00	144 27	6 123	54.3	32.69
MobileNetV2	0.926	0.840	0.881	88.66	126 10	24 140	2.2	7.05
NASNetMobile	0.891	0.980	0.933	93.00	147 18	3 132	4.2	40.65
ResNetV2_101	0.728	0.947	0.823	79.66	142 53	8 97	44.5	19.09
VGG16	0.672	0.807	0.733	70.66	121 59	29 91	134.2	7.73
VGG19	0.835	0.573	0.680	73.00	86 17	64 133	139.5	9.34
LesionScanNet	0.987	0.993	0.990	99.00	149 2	1 148	0.765	6.80

Best values are given in bold

The LesionScanNet model obviously outperforms the benchmark models in terms of all evaluation metrics. There are only two false positive and one false negative predictions yielding an accuracy of 99%. These misclassified samples are further inspected in Sect. “Discussion”. It is also worth mentioning that the training for LesionScanNet was terminated by early stopping at epoch 96.

Regarding computational cost, in addition to having the lowest number of parameters at 0.765 million compared to state-of-the-art models, the LesionScanNet model also achieved the lowest inference time of 6.80 ms per image, which is slightly better than MobileNetV2. However, the number of parameters is nearly three times lower than those of MobileNetV2.

Ablation study

The ablation study seeks to determine the impact of 1×1 kernel branch in the proposed model. This was done by eliminating the 1×1 kernel branch in all blocks of the model, leaving the 3×3 kernel branch unchanged. The model in the ablation study was trained using the same optimizer and hyperparameter settings already used in the full model. The impact of this change on the classification performance is provided in Table 6. Even though the number of false positives decreased, there is degradation in accuracy, recall, and F1-score values because of the increased false negative predictions. Overall, the achieved accuracy is 97.33% with one false positive and

Table 6 The results of ablation study

Precision	Recall	F1-score	Prediction accuracy	Confusion matrix	
0.993	0.953	0.973	97.33%	143 1	7 149

seven false negative predictions. In addition, the best validation results obtained at epoch 155 indicating that the full LesionScanNet model is capable of learning in smaller number of epochs.

Generalization study

In this part of the experiments, the scalability of LesionScanNet has been investigated through assessing its performance on a different type of medical image dataset. For this purpose, a publicly available chest X-ray image dataset has been used [30]. This dataset comprises 9207 chest X-ray images, categorized as normal (N, 3269 images), COVID-19 (C, 1281 images), bacterial pneumonia (B, 3001 images) and viral pneumonia (V, 1656 images). LesionScanNet has been benchmarked against a *sequential model* which is reported to achieve the best performance for this dataset in the literature [3].

The generalization study followed the identical training settings, including the optimizer function, data augmentation steps, and a training/validation/test set split

of 60%, 20%, and 20%, respectively, with the sequential model to ensure a fair comparison. This corresponds to a test set of 1841 images on which the performance metrics are calculated. According to the comparative results, LesionScanNet is able to achieve a predictive accuracy of 90.22% while the accuracy value for the other model is 89.89%. The results are detailed in Tables 7 and 8 providing the confusion matrices for both models. Based on Table 8, LesionScanNet outperforms the sequential model in the classification of normal samples, COVID-19, and viral pneumonia scans. On the other hand, the sequential model is better in detecting bacterial pneumonia. For detailed analysis, the same metrics in Sect. “Performance evaluation metrics” have been provided in Table 9. These tables present the confusion matrix and the precision, recall, and F1-score values for both models and all four classes in the dataset, respectively.

Discussion

This study evaluates the performance of a novel CNN model, LesionScanNet, on detection of acute appendicitis cases from CT images. It also compares the achieved results with those obtained by nine benchmark models in the literature. The experiments within the scope of the study have been detailed by providing an ablation study as well as a generalization study. Detailed discussions about the associated results are provided in the following subsections.

Performance comparison of LesionScanNet and benchmark models

The results show that the LesionScanNet model can predict acute appendicitis with the highest accuracy of 99% and significantly outperforms the best result obtained by pre-trained models, among which NASNetMobile achieved the highest accuracy of 93%. In addition, the model has a 98.79% and 99.30% recall and precision rating, respectively, which results in an F1-score of 99%.

Table 7 The confusion matrix presenting the classes of X-ray images

	Predicted			
	Covid-19 (C)	Normal (N)	Pneumonia bacterial (B)	Pneumonia viral (V)
Actual				
Covid-19 (C)	P_{CC}	P_{NC}	P_{BC}	P_{VC}
Normal (N)	P_{CN}	P_{NN}	P_{BN}	P_{VN}
Pneumonia bacterial (B)	P_{CB}	P_{NB}	P_{BB}	P_{VB}
Pneumonia viral (V)	P_{CV}	P_{NV}	P_{BV}	P_{VV}

Table 8 Comparison of the confusion matrix between LesionScanNet model and the sequential model

	LesionScanNet model				Sequential model			
Confusion matrix	254	2	0	0	247	1	7	1
	0	646	3	5	1	640	3	10
	4	14	533	49	2	8	555	35
	0	11	92	228	1	10	107	213

Table 9 Comparison of the results between the LesionScanNet model and the sequential model

	LesionScanNet			Sequential model		
	Precision	Recall	F1-score	Precision	Recall	F1-score
Covid-19	0.984	0.992	0.988	0.984	0.965	0.974
Normal	0.960	0.988	0.974	0.971	0.979	0.975
Pneumonia Bacterial	0.849	0.888	0.868	0.826	0.925	0.873
Pneumonia Viral	0.809	0.689	0.744	0.822	0.644	0.723

Best values are given in bold

This situation shows a balanced and high performance in detecting both classes. This balance is important in medical diagnosis, where incorrect classification, especially false negatives, carries serious consequences for patients.

MobileNetV2 and VGG19 are the models possessing the smallest and highest numbers of parameters among the benchmark models, respectively. However, it is not possible to observe a direct relationship between the model performance and the number of parameters because the accuracy value of VGG19 is much lower than MobileNetV2. The reason for this situation may be the poor feature extraction capability of the related models, particularly when it comes to medical diagnostic problems such as identifying acute appendicitis. This is addressed by LesionScanNet which outperforms the benchmark models by being a lighter model with fewer parameters, which is 0.765M parameters, and achieves the lowest inference time of 6.80 ms per image. This implies that the proposed model has only one-third of the parameters contained in the MobileNetV2 model.

Note that there are only three misclassified samples associated with LesionScanNet predictions. For further consideration, these misclassifications were inspected manually to understand the reasons for this situation (Fig. 3). It was observed that extra challenges may be introduced due to fluid density and calcifications around the lesions or small anatomical structures of the lesions.

In the false negative prediction, some calcifications and fluid densities causing an infection were spotted around the acute appendicitis region. The infection area as well as the acute appendicitis area are labelled by red and blue boxes in Fig. 3a. The density observed in the upper part is likely due to the opening of the lumen of the appendix and the expulsion of hardened feces, resulting in this appearance. On the other hand, the false positive predictions are associated with the very tiny appearance of the appendix vermiformis (Fig. 3b). In particular, the small lumen diameter of the

marked extension showing thin intestine attachment (the upper part of the blue mark in Fig. 3c). This situation is accompanied by a double wall appearance that is seen in the pathology of acute appendicitis.

Robustness of the LesionScanNet model

The robustness of LesionScanNet has been demonstrated through ablation and generalization studies. The ablation study aimed to highlight the importance of the branch with 1×1 kernels in the DKB structure. Therefore, it demonstrated how the omission of the branch altered the predictive abilities of the model. As a result of the ablation study, the accuracy value dropped from 99 to 97.33% when the branch associated with kernel size of 1×1 is removed and the model needed more epochs to achieve its highest accuracy. The recall and F1-score for acute appendicitis detection was lower than the full model without ablation. This means that the model, after removing the 1×1 kernel branch, was less effective in extracting the relevant features, and hence, in detecting acute appendicitis cases. A critical point is that the highest predictive accuracy was achieved after 155 epochs in the modified model, whereas it took 96 epochs to reach this level in the full model. This shows the relevance of the 1×1 kernel branch in feature extraction at a channel level and its applicability in classifying acute appendicitis cases. This situation emphasizes the role of the 1×1 branch in enhancing the model performance as well as in speeding up the process of training through capturing of the significant features.

Within the generalization study, the model was evaluated on a different detection problem (classification of lung diseases) using a different type of medical images (X-ray images). LesionScanNet demonstrated high accuracy, outperforming the best results on the same problem in the literature. In particular, LesionScanNet performed better in terms of recall in three out of four categories, which makes it highly sensitive and capable of effectively detecting most true positives. This is especially



Fig. 3 False negative (a) and false positive (b, c) predictions for LesionScanNet

the case with pneumonia viral class, where the recall is greater than that of the sequential model. Higher recall value means that the LesionScanNet model is less prone to have false negatives for pneumonia viral, which is very important in medical diagnostics.

The sequential model, on the other hand, although exhibiting relatively lower recall in the COVID-19 and pneumonia viral classes, has a slightly better performance when it comes to precision for the normal and pneumonia viral types. Nevertheless, LesionScanNet achieves slightly better precision-recall trade-off for COVID-19 and pneumonia viral classifications in terms of the F1-scores. It is notable that both models perform poorly in detection of pneumonia viral class when compared to other three classes. In summary, the LesionScanNet model seems to have more balanced performance with the sensitivity slightly skewed towards higher values for most of the classes, which can be useful in medical applications where false negatives have higher burden than false positives.

Conclusion

In this study, a novel convolutional neural network model, namely LesionScanNet, has been proposed for detection of acute appendicitis in CT images. LesionScanNet comprises five unique processing blocks known as DualKernel blocks each of which includes two branches to handle the input data. Both branches use a set of convolutional, expansion, and separable convolutional layers. Meanwhile, one branch applies a kernel of size 1×1 whereas the other applies a kernel of size 3×3 . This model is unique in the sense that it is the first to use a 1×1 kernel in the medical domain that only extracts features at the channel level along the entire processing pipeline in the model, a technique which is specifically important in medical image processing. The developed model has shown an outstanding performance in acute appendicitis classification with a prediction accuracy of 99%, corresponding to only three misclassified samples in the test set. The performance of LesionScanNet has been compared with some benchmark models and an obvious outperformance has been observed. This points out the possibility of employing the proposed model as an auxiliary tool in the medical diagnosis of acute appendicitis. In addition to its high performance, LesionScanNet is lightweight, possessing a small number of parameters not more than 0.765M which is much less compared to the number of parameters in the lightest pre-trained benchmark models used in this study, such as the MobileNetV2 model. Additionally, the inference time is 6.80 ms per image, making it the lightest model compared with benchmark models. This means that it can be deployed on hardware with limited computational resources.

The proposed model can be evaluated using different medical datasets like ultrasound images or 3D images as a future work. Moreover, the experiments may be furthered after the dataset is expanded by involving images with more diverse acute appendicitis appearances. Another future work option can be investigating whether there is a trade-off between the number of model parameters and the classification performance.

Declarations

Conflict of interest

There are no competing interests and no funding related to this work.

Ethical approval

The study is approved by the ethical committee of the Kanuni Sultan Suleyman Research and Training Hospital (Approval Date: 11/07/2024, Reference No: 2024.07.157).

Consent to participate

The need for informed consent is waived by the committee due to the retrospective nature of archived datasets and fully anonymized personal information.

Author details

¹Electrical and Electronics Engineering Department, Çukurova University, 01330 Adana, Turkey. ²Biomedical Engineering Department, Çukurova University, 01330 Adana, Turkey. ³General Surgery Department, Derik State Hospital, 47800 Mardin, Turkey. ⁴General Surgery Department, Kanuni Sultan Suleyman Research and Training Hospital, 34303 Istanbul, Turkey. ⁵General Surgery Department, Bilkent City Hospital, 06800 Ankara, Turkey. ⁶Interventional Radiology Department, Kanuni Sultan Suleyman Research and Training Hospital, 34303 Istanbul, Turkey. ⁷Section for Fisheries Technology, Institute of Aquatic Resources, DTU Aqua, Technical University of Denmark, 9850 Hirtshals, Denmark.

Received: 23 August 2024 Accepted: 19 November 2024

Published: 7 December 2024

References

- Gao F, et al. SD-CNN: a shallow-deep CNN for improved breast cancer diagnosis. *Comput Med Imaging Graph*. 2018;70:53–62.
- Tripathi S, et al. Lung disease detection using deep learning. *Int J Innov Technol Explor Eng*. 2021;10(8):1–10.
- Hariri M, Avşar E. COVID-19 and pneumonia diagnosis from chest X-ray images using convolutional neural networks. *Netw Model Anal Health Inform Bioinform*. 2023;12(1):17.
- Sharif M, et al. Deep CNN and geometric features-based gastrointestinal tract diseases detection and classification from wireless capsule endoscopy images. *J Exp Theor Artif Intell*. 2021;33(4):577–99.
- Al WA, Yun ID, Lee KJ. Reinforcement learning-based automatic diagnosis of acute appendicitis in abdominal ct. 2019. arXiv preprint [arXiv:1909.00617](https://arxiv.org/abs/1909.00617).
- Chollet F. Xception: deep learning with depthwise separable convolutions. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2017.
- Rajpurkar P, et al. AppendixNet: deep learning for diagnosis of appendicitis from a small dataset of CT exams using video pretraining. *Sci Rep*. 2020;10(1):3958.
- Park JJ, et al. Convolutional-neural-network-based diagnosis of appendicitis via CT scans in patients with acute abdominal pain presenting in the emergency department. *Sci Rep*. 2020;10(1):9556.
- Lee GP, et al. Enhancing disease classification in abdominal CT scans through RGB superposition methods and 2D convolutional neural

- networks: a study of appendicitis and diverticulitis. *Comput Math Methods Med*. 2023. <https://doi.org/10.1155/2023/7714483>.
10. Marcinkevičs R, et al. Interpretable and intervenable ultrasonography-based machine learning models for pediatric appendicitis. *Med Image Anal*. 2024;91:103042.
 11. Mijwil MM, Aggarwal K. A diagnostic testing for people with appendicitis using machine learning techniques. *Multimed Tools Appl*. 2022;81(5):7011–23.
 12. Küçükakçali Z, Akbulut S, Çolak C. Value of fecal calprotectin in prediction of acute appendicitis based on a proposed model of machine learning. *Türk J Trauma Emerg Surg*. 2023;29(6):655.
 13. Marcinkevičs R, et al. Using machine learning to predict the diagnosis, management and severity of pediatric appendicitis. *Front Pediatr*. 2021;9:662183.
 14. Akmeshe OF, et al. The use of machine learning approaches for the diagnosis of acute appendicitis. *Emerg Med Int*. 2020. <https://doi.org/10.1155/2020/7306435>.
 15. Byun J, Park S, Hwang SM. Diagnostic algorithm based on machine learning to predict complicated appendicitis in children using CT, laboratory, and clinical features. *Diagnostics*. 2023;13(5):923.
 16. Singh D, et al. Assessment of machine learning-based medical directives to expedite care in pediatric emergency medicine. *JAMA Netw Open*. 2022;5(3):e222599–e222599.
 17. Lin H-A, Lin L-T, Lin S-F. Application of artificial neural network models to differentiate between complicated and uncomplicated acute appendicitis. *J Med Syst*. 2023;47(1):38.
 18. Goswami R, et al. Comparison and analysis of machine learning techniques for the prediction of acute appendicitis. *J Appl Inf Sci*. 2020;8(1&2):14–21.
 19. Avcı E. Effects of image preprocessing on the performance of convolutional neural networks for pneumonia detection. In: 2021 international conference on INnovations in Intelligent SysTems and Applications (INISTA). IEEE; 2021.
 20. Lu H-C, Loh E-W, Huang S-C. The classification of mammogram using convolutional neural network with specific image preprocessing for breast cancer detection. In: 2019 2nd International Conference on Artificial Intelligence and Big Data (ICAIBD). IEEE; 2019.
 21. Tan M, Le Q. Efficientnet: Rethinking model scaling for convolutional neural networks. In: International conference on machine learning. PMLR; 2019.
 22. Szegedy C et al. Rethinking the inception architecture for computer vision. In: Proceedings of the IEEE conference on computer vision and pattern recognition. 2016.
 23. Szegedy C et al. Inception-v4, inception-resnet and the impact of residual connections on learning. In: Proceedings of the AAAI conference on artificial intelligence. 2017.
 24. Sandler M et al. Mobilenetv2: Inverted residuals and linear bottlenecks. In: Proceedings of the IEEE conference on computer vision and pattern recognition. 2018.
 25. Zoph B et al. Learning transferable architectures for scalable image recognition. In: Proceedings of the IEEE conference on computer vision and pattern recognition. 2018.
 26. He K et al. Deep residual learning for image recognition. In: Proceedings of the IEEE conference on computer vision and pattern recognition. 2016.
 27. Simonyan K, Zisserman A. Very deep convolutional networks for large-scale image recognition. 2014. arXiv preprint [arXiv:1409.1556](https://arxiv.org/abs/1409.1556).
 28. Russakovsky O, et al. Imagenet large scale visual recognition challenge. *Int J Comput Vision*. 2015;115:211–52.
 29. Kingma DP, Ba J. Adam: A method for stochastic optimization. 2014. arXiv preprint [arXiv:1412.6980](https://arxiv.org/abs/1412.6980).
 30. Sait U, et al. Curated dataset for COVID-19 posterior-anterior chest radiography images (X-rays). Mendeley Data. 2020;1:1. <https://doi.org/10.17632/9xkhgts2s6.3>.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.